ML Visualisation Tool — Sprint 2 Screenshot Report

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GitHub repo: `https://github.com/MuffinMufido/HealthML.wiki`

Live demo URL: `http://localhost:5173/`

This PDF demonstrates that Steps 1–3 are functional via labelled screenshots and brief explanations.

Step 1 — Clinical Context

Step 1A: Specialty selection (Cardiology)

ML Clinical Visualizer
For Healthcare Professionals

1 Clinical Context — 2 Data Exploration — 3 Data Preparation — 4 Model & Parameters — 5 Results — 6 Explainability — 7 Ethics & Bias

Step 1: Clinical Context

Choose a medical specialty to explore how AI can assist in that clinical area. Each specialty has a real-world problem where machine learning can support clinical decision-making.

No technical background needed. This tool uses plain language to explain every concept. Hover over highlighted terms for definitions, and follow each step at your own pace.

- **Cardiology** — 30-Day Readmission Risk
- **Radiology** — Pneumonia Classification
- **Nephrology** — Chronic Kidney Disease Stage
- **Oncology — Breast** — Breast Biopsy Malignancy
- **Neurology — Parkinson's** — Parkinson's Disease Detection
- **Endocrinology — Diabetes** — Diabetes Onset Prediction
- **Hepatology — Liver** — Liver Disease Detection
- **Cardiology — Stroke** — Stroke Risk Prediction
- **Mental Health** — Depression Severity
- **Pulmonology — COPD** — COPD Exacerbation Risk
- **Haematology — Anaemia** — Anaemia Type Classification
- **Dermatology** — Skin Lesion Classification
- **Ophthalmology** — Diabetic Retinopathy Severity
- **Orthopaedics — Spine** — Disc Herniation Detection
- **ICU / Sepsis** — Sepsis Onset Prediction
- **Obstetrics — Fetal Health** — Fetal Health Classification
- **Cardiology — Arrhythmia** — Cardiac Arrhythmia Detection
- **Oncology — Cervical** — Cervical Cancer Risk
- **Thyroid / Endocrinology** — Thyroid Function Classification
- **Pharmacy — Readmission** — Hospital Readmission Risk

Question: 30-Day Readmission Risk

CLINICAL CONTEXT

This model utilizes Heart Failure Clinical Records to predict the 30-day readmission risk after a heart failure discharge. Predicting these outcomes helps target post-discharge interventions and prevents adverse events.

TARGET VARIABLE: DEATH_EVENT (binary)

What ML cannot do: It cannot replace your clinical judgment. It can flag high-risk patients, but you make the final decision.

Continue to Data Exploration →

What Will Be Produced in Each Step

Step	What You Will Create	Plain English Meaning
1	Clinical Brief	The problem we are solving, and the safety rules
2	Data Profile	Understanding your patient dataset — who is in it and what is missing
3	Preprocessing Recipe	Cleaning and preparing data so the model can learn correctly
4	Trained Model	A computer programme that has learned patterns from past patients
5	Evaluation Report	How accurately does the model perform predictions?
6	Explanation	Why did the model flag a specific patient as high risk?
7	Ethics Checklist	Is the model fair for all patient groups? Who oversees it?

Remember: A human doctor or nurse must always review the model's suggestions. This tool helps you learn — it does not make clinical decisions.

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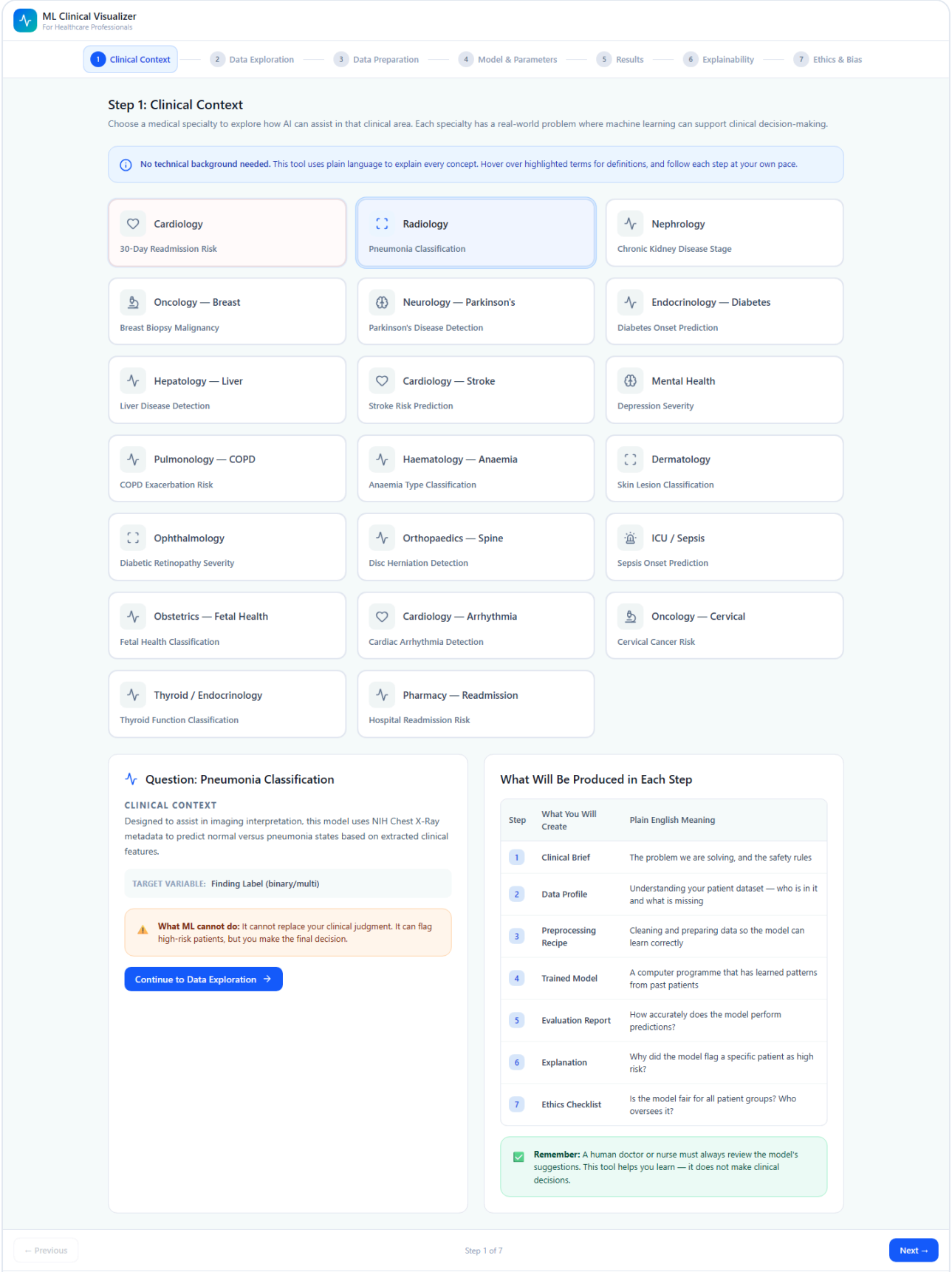
Screenshot: step1/domain_cardiology_selected.png

Selecting the Cardiology specialty updates the clinical context panel (disease/problem, population context, and prediction target), proving domain-specific Step 1 content renders correctly.

Step 1A: Specialty selection (Radiology)



Screenshot: step1/domain_radiology_selected.png

Selecting Radiology switches the clinical context text and target variable to the Radiology domain, demonstrating that different specialty buttons produce correct domain-specific content.

Step 1A: Specialty selection (Nephrology)

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- Oncology — Cervical — Cervical Cancer Risk
- Thyroid / Endocrinology — Thyroid Function Classification
- Pharmacy — Readmission — Hospital Readmission Risk

Question: Chronic Kidney Disease Stage

CLINICAL CONTEXT

Predicts chronic kidney disease stage from routine lab values. Early detection helps slow disease progression. Data Source: UCI CKD Dataset (400 patients).

TARGET VARIABLE: classification (CKD / not CKD)

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Continue to Data Exploration →

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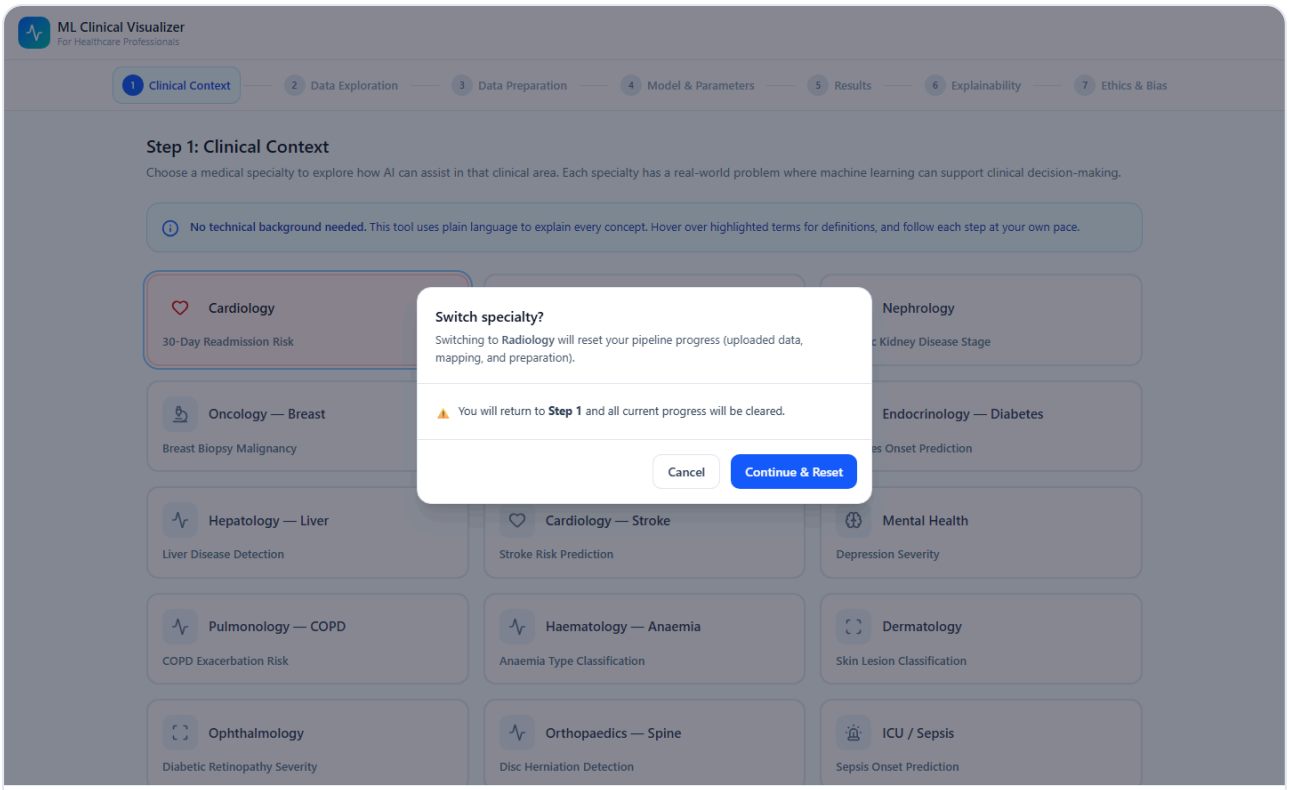
Screenshot: step1/domain_nephrology_selected.png

Selecting Nephrology updates the clinical context (disease/problem + target) for that specialty, completing the “at least 3 specialties selected” Step 1 requirement.

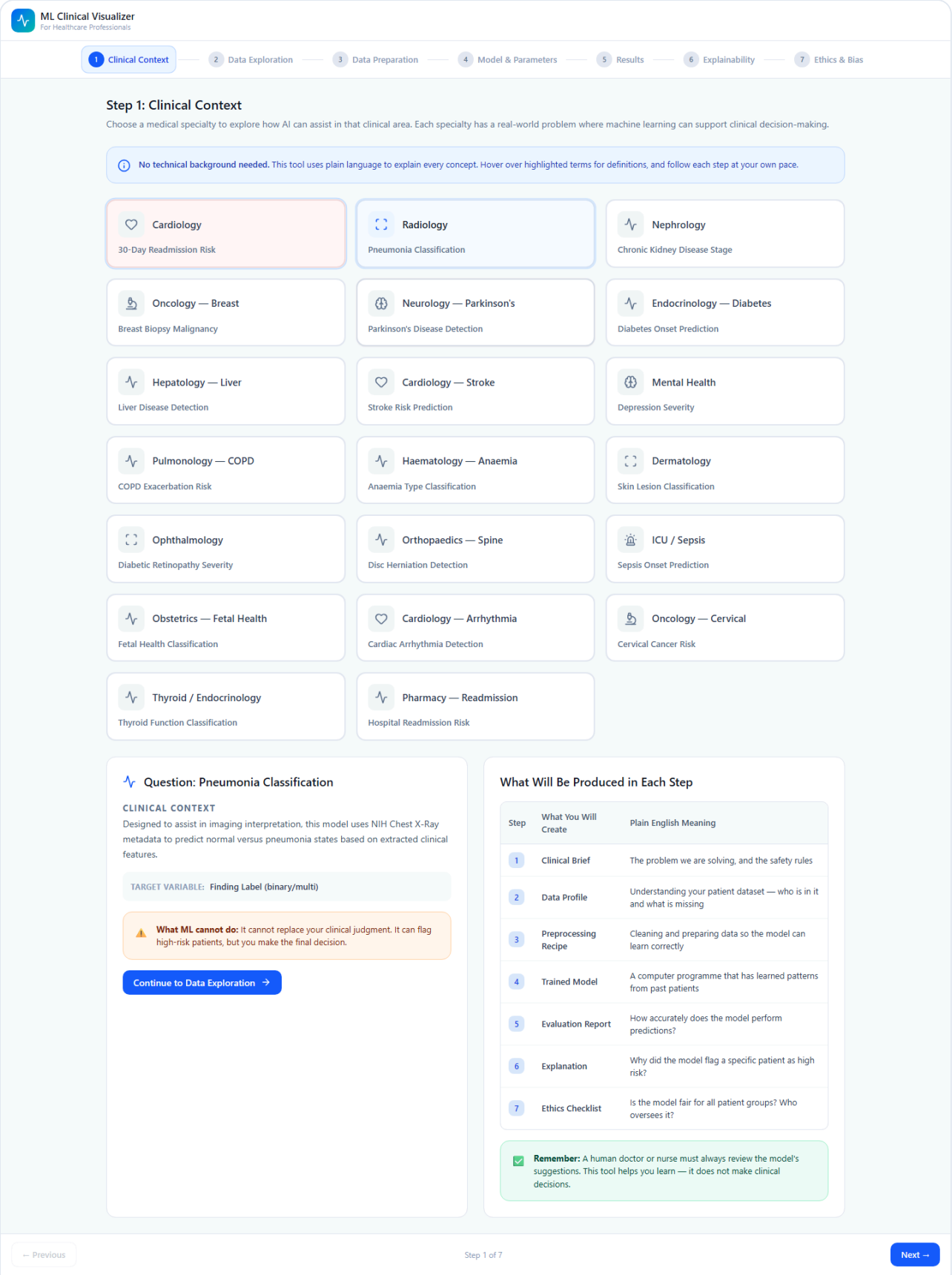
Step 1B: Pipeline reset behaviour (confirmation shown)



Screenshot: step1/pipeline_reset_confirmation.png

After making progress, switching specialties triggers a reset confirmation modal. This proves the required “pipeline reset behaviour when switching domains” is enforced before changing context.

Step 1B: Pipeline reset behaviour (after reset)

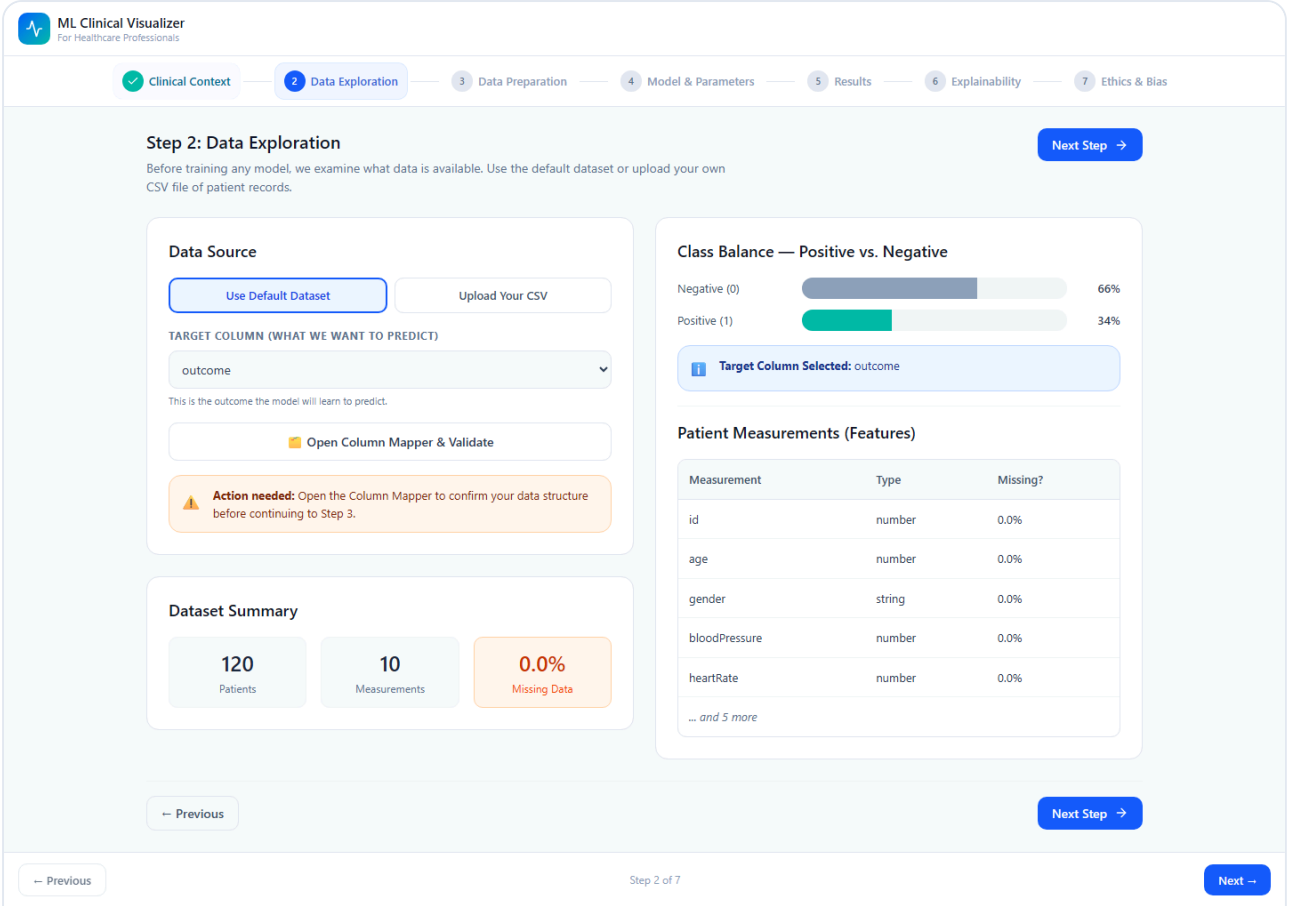


Screenshot: step1/pipeline_reset_completed.png

After confirming, the pipeline resets back to Step 1 and the new specialty becomes active (prior dataset/mapping/prep progress cleared), completing the reset flow demonstration.

Step 2 — Data Exploration & Column Mapper

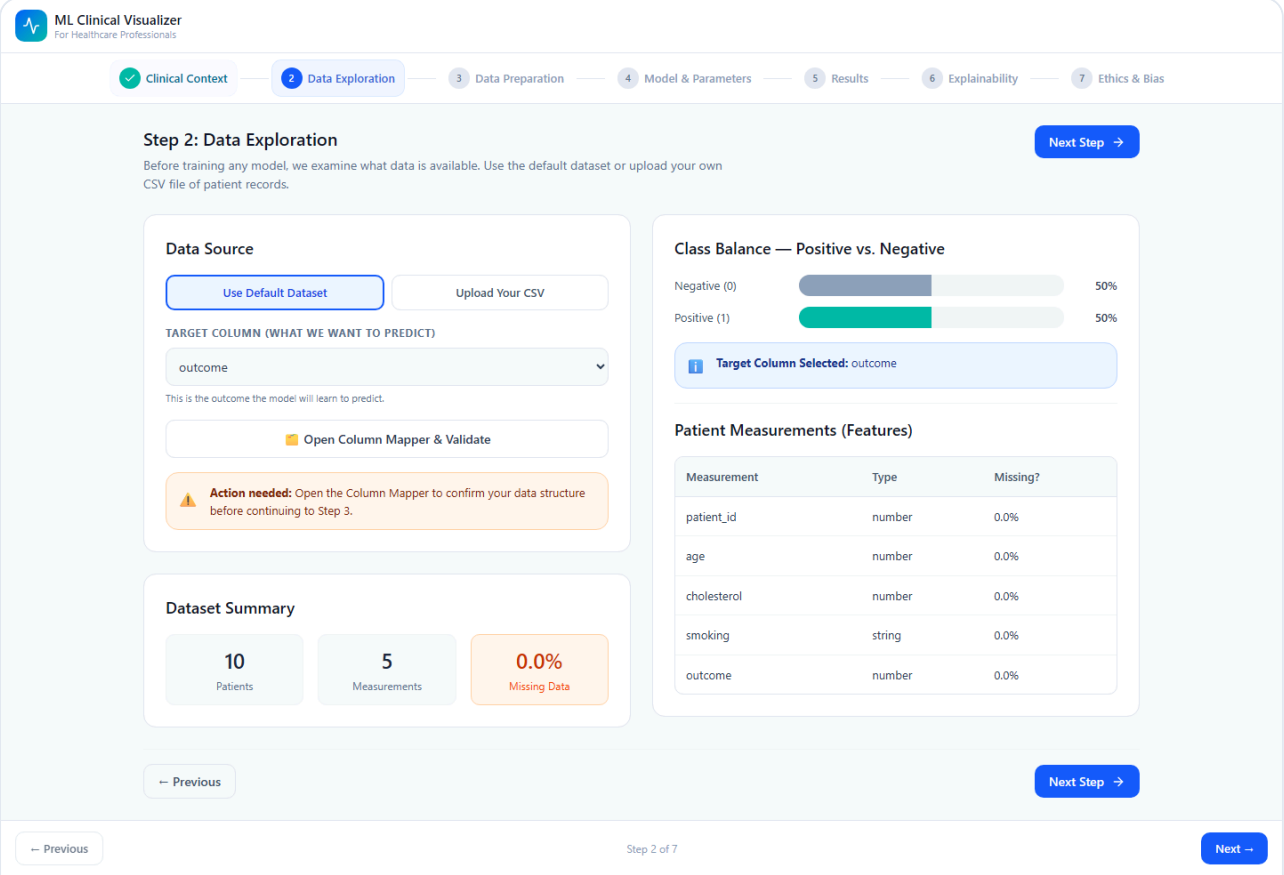
Step 2A: Built-in example dataset loaded



Screenshot: step2/example_dataset_loaded.png

The built-in example dataset loads successfully and shows dataset summary, class distribution, and column summaries—demonstrating the Step 2 example-data feature works end-to-end.

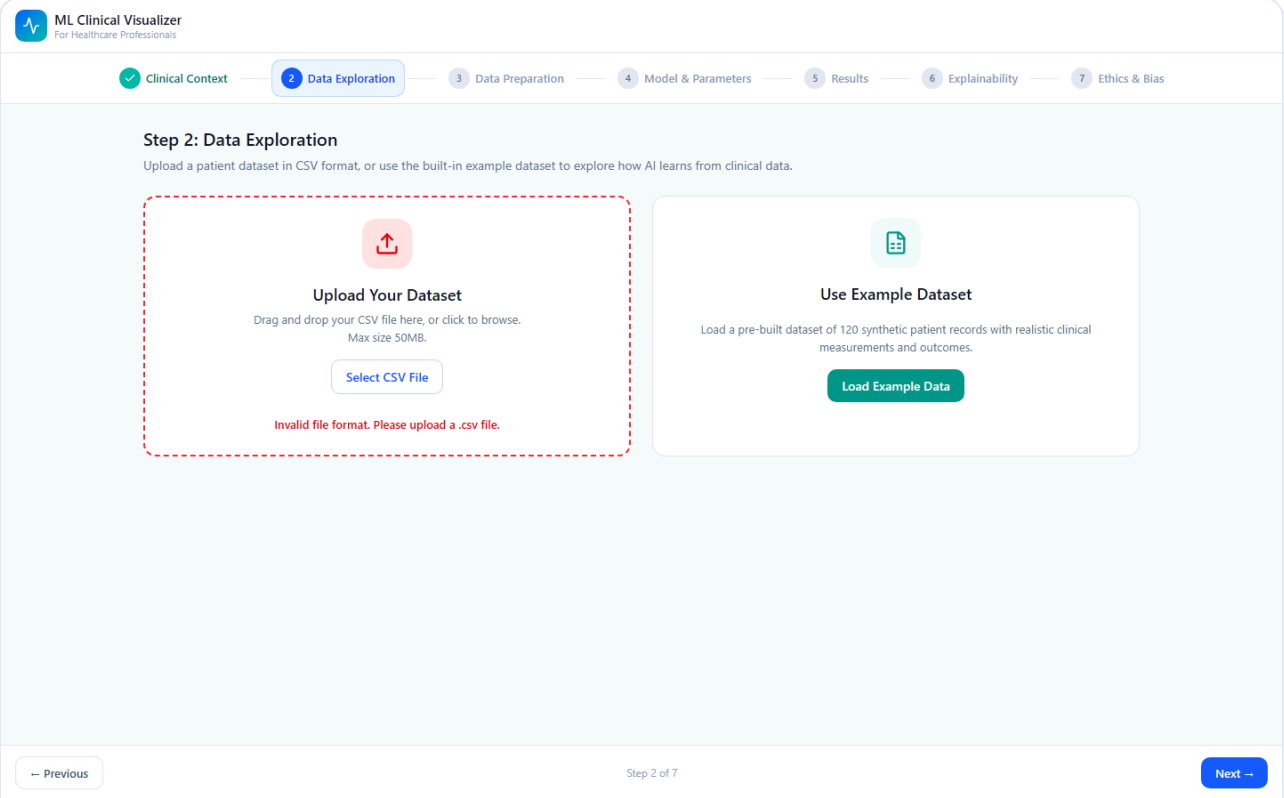
Step 2B: CSV upload — valid file accepted



Screenshot: csv/upload_valid_v1_success.png

Uploading a valid CSV is accepted and renders the dataset summary view, proving the CSV upload pipeline works correctly.

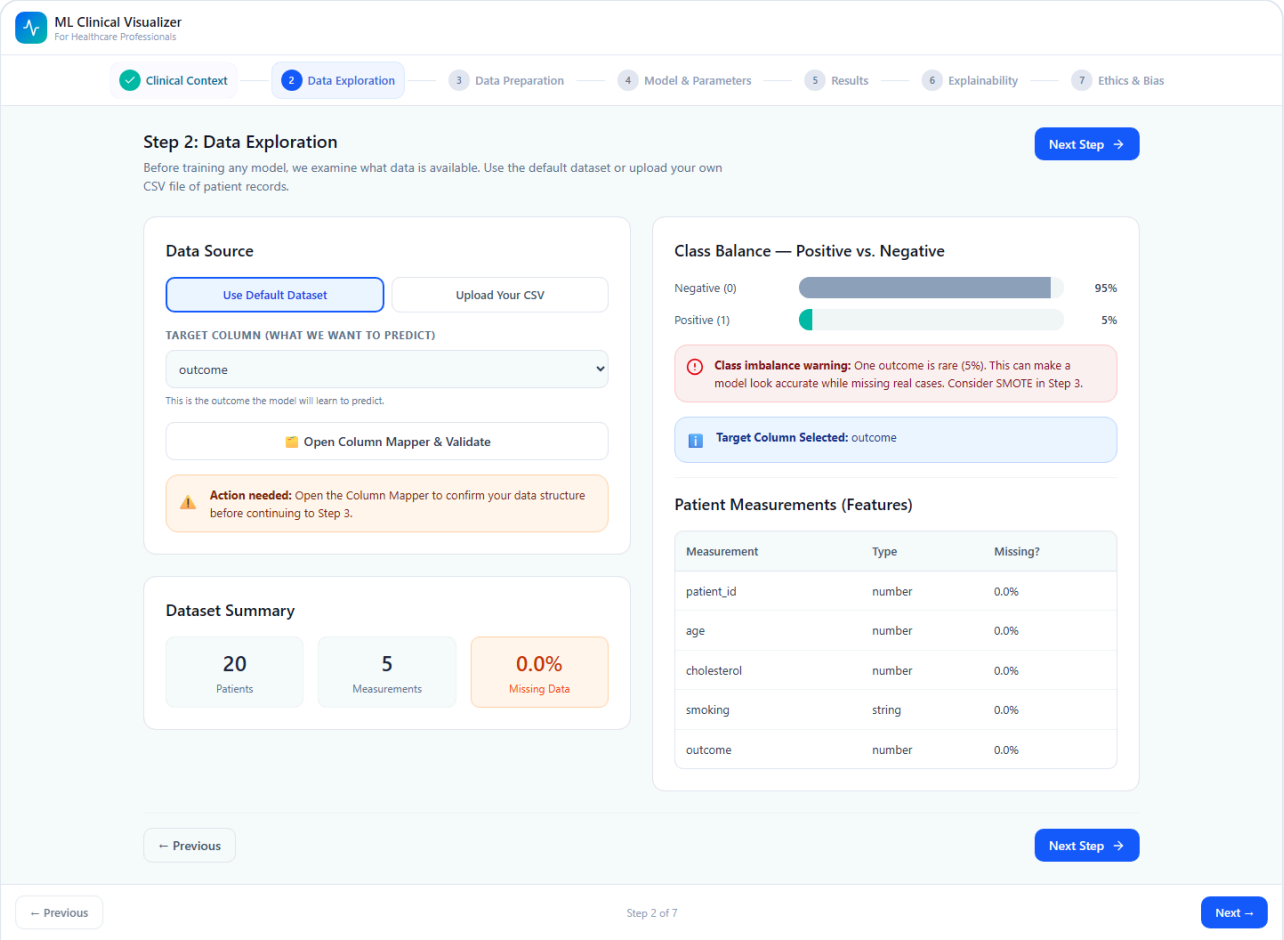
Step 2B: CSV upload — invalid file rejected (friendly error)



Screenshot: csv/upload_invalid_not_csv_blocked.png

Uploading an invalid (non-CSV) file is rejected with a clear, user-friendly error message, demonstrating robust input validation in Step 2.

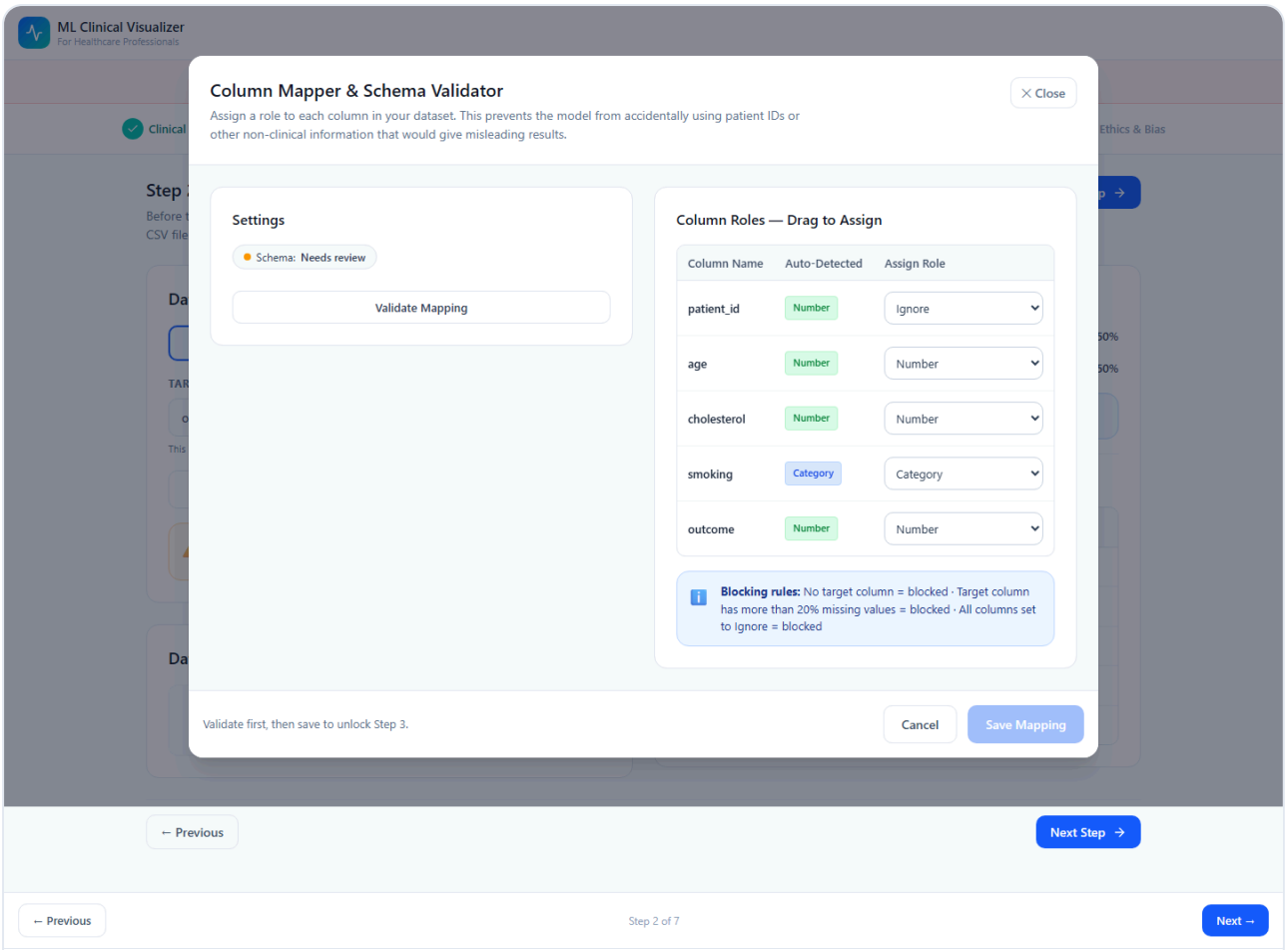
Step 2C: Class imbalance warning (when applicable)



Screenshot: step2/imbalance_warning_and_summaries.png

When the outcome class is highly imbalanced, the tool displays a class-imbalance warning alongside class distribution and column summaries, matching the Step 2 requirement.

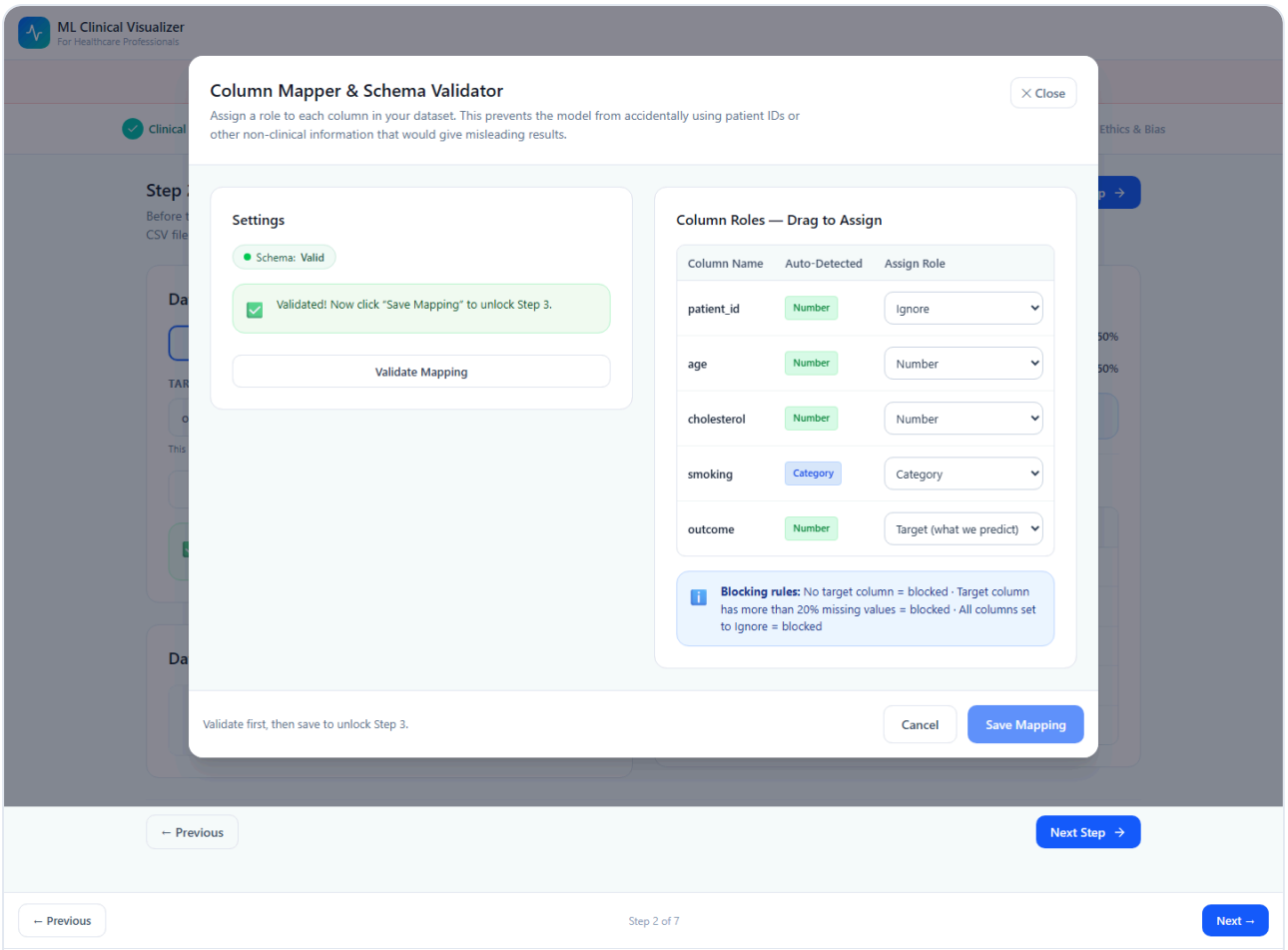
Step 2D: Column Mapper opened



Screenshot: step2/column_mapper_open.png

The Column Mapper modal opens and allows feature/target role assignment, enabling schema definition prior to Step 3.

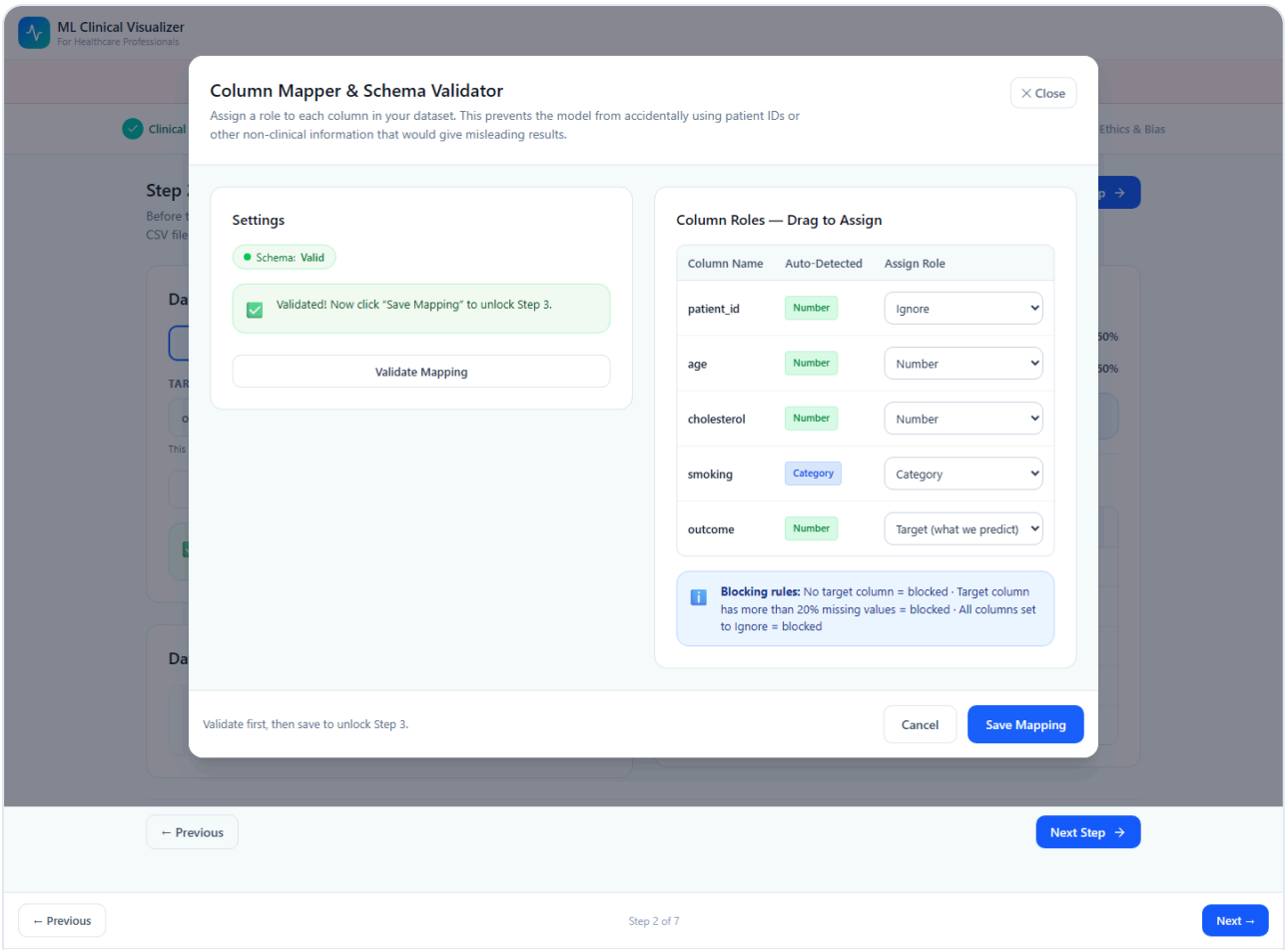
Step 2D: Validate mapping



Screenshot: step2/column_mapper_validated_message.png

Clicking "Validate Mapping" checks schema rules (e.g., exactly one target) and shows a validation success message—Step 2 validate stage completed.

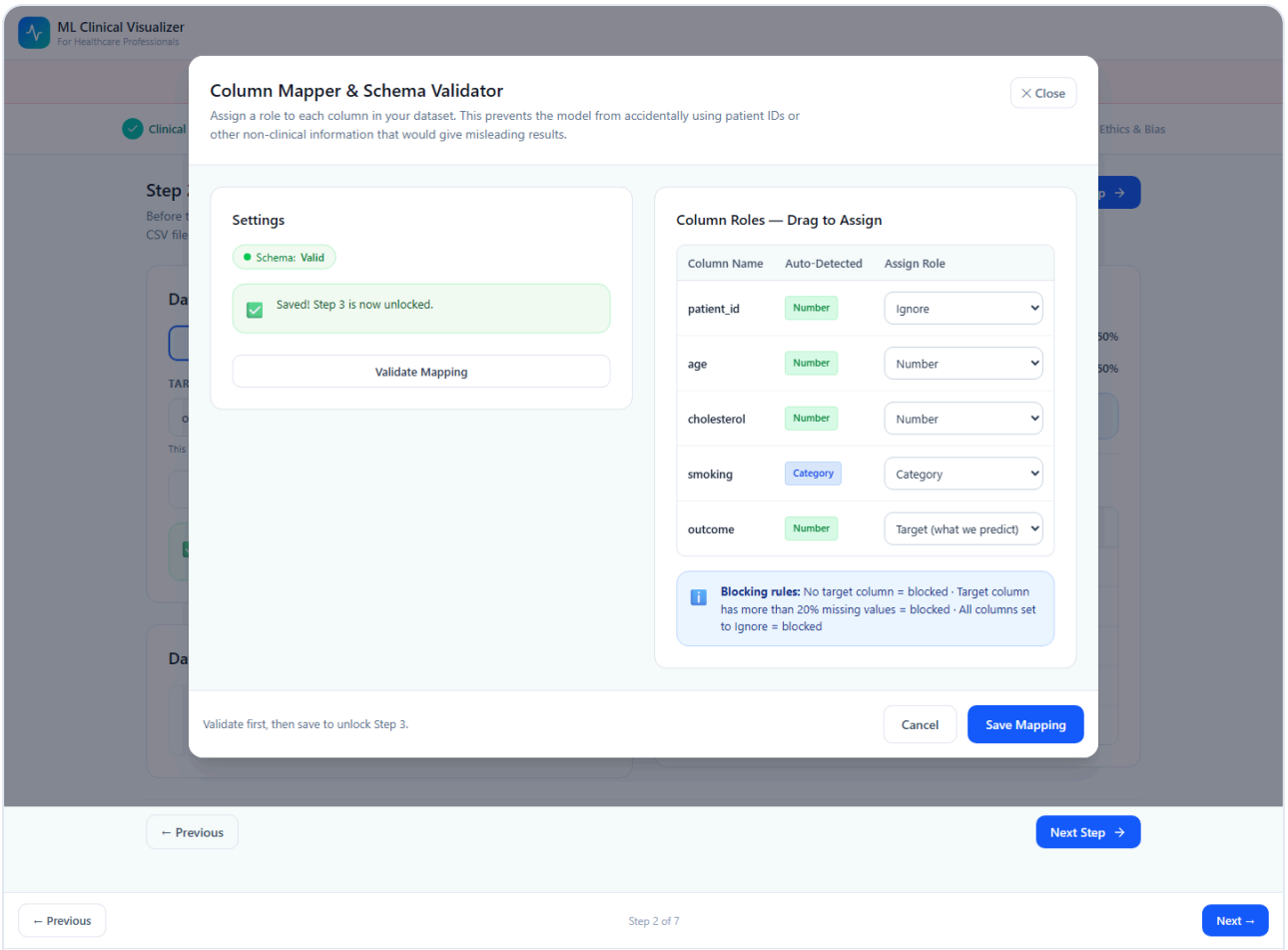
Step 2D: Save enabled after validation



Screenshot: step2/column_mapper_save_enabled.png

After validation succeeds, “Save Mapping” becomes enabled, enforcing the required validate → save flow (cannot save before validation).

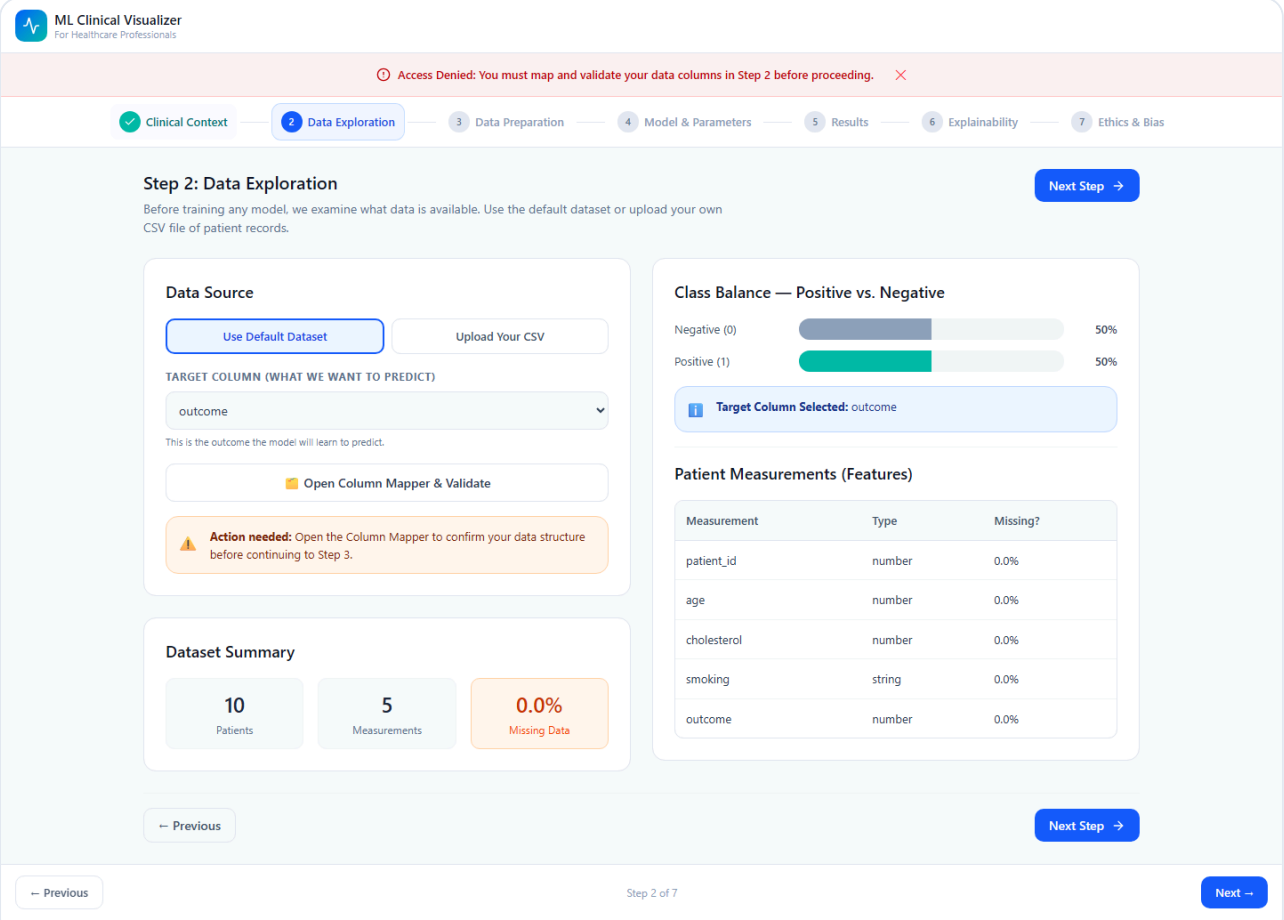
Step 2D: Save completed (Step 3 unlocked)



Screenshot: step2/column_mapper_saved_success.png

Saving the mapping completes validate → save and unlocks Step 3 (schema gate satisfied), as confirmed by the success message in the modal.

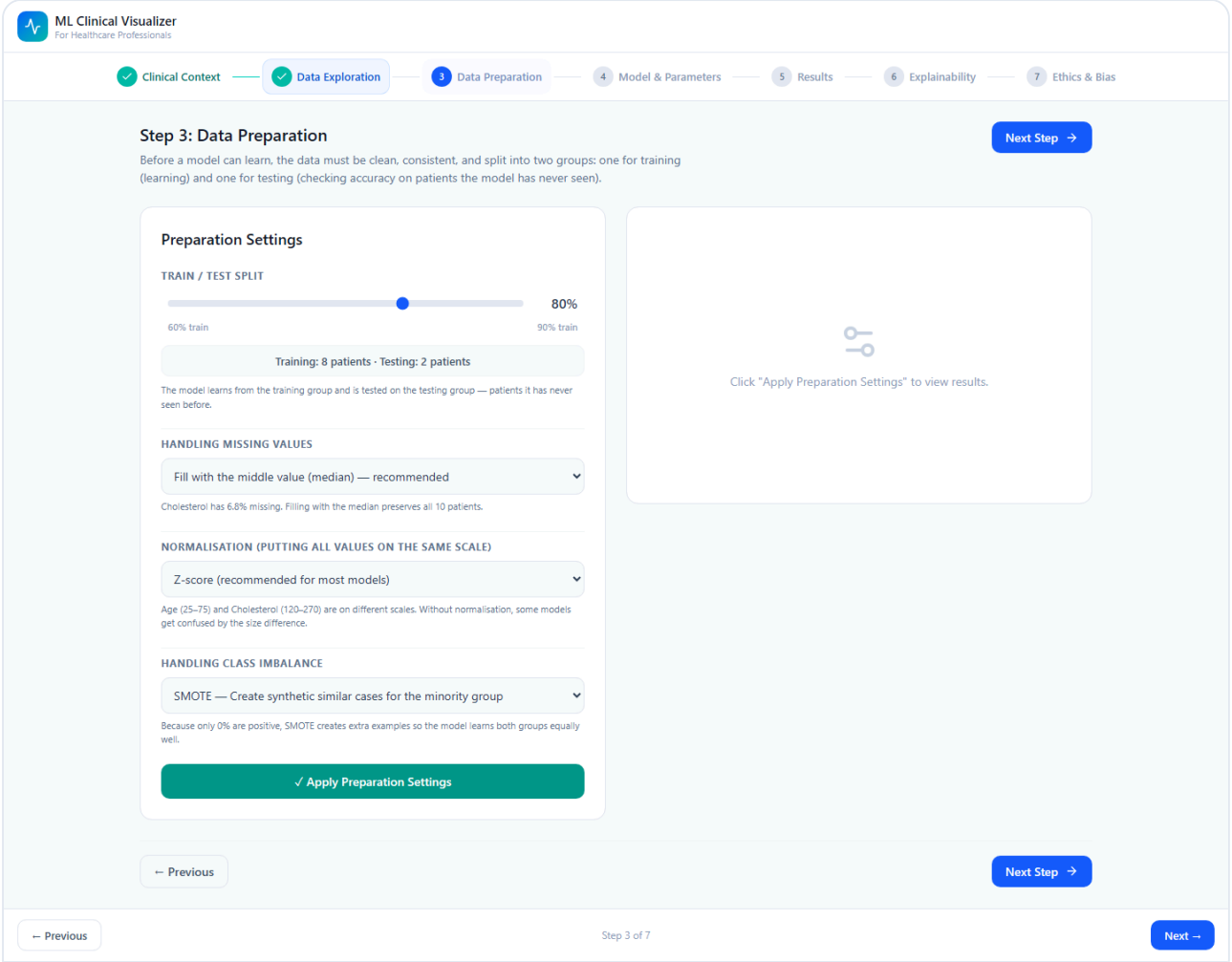
Step 2E: Gate control — blocked before save



Screenshot: gates/step3_blocked_before_save.png

Attempting to access Step 3 before saving Column Mapper shows an Access Denied/blocked banner, proving the Step 3 gate is enforced.

Step 2F: Gate control — accessible after save

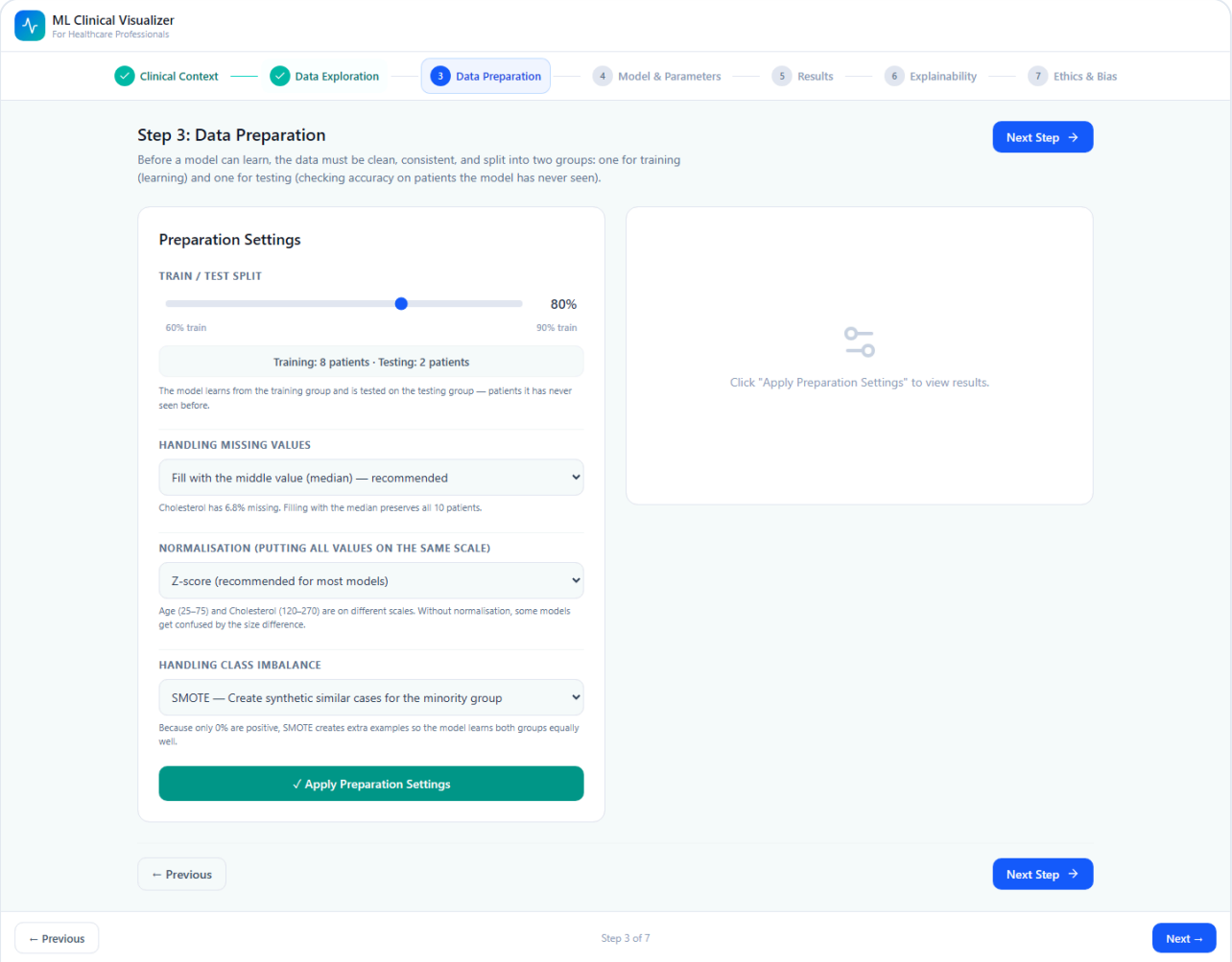


Screenshot: gates/step3_accessible_after_save.png

After saving Column Mapper, Step 3 becomes accessible without the blocked banner, demonstrating the gate opens only after the required action.

Step 3 — Data Preparation

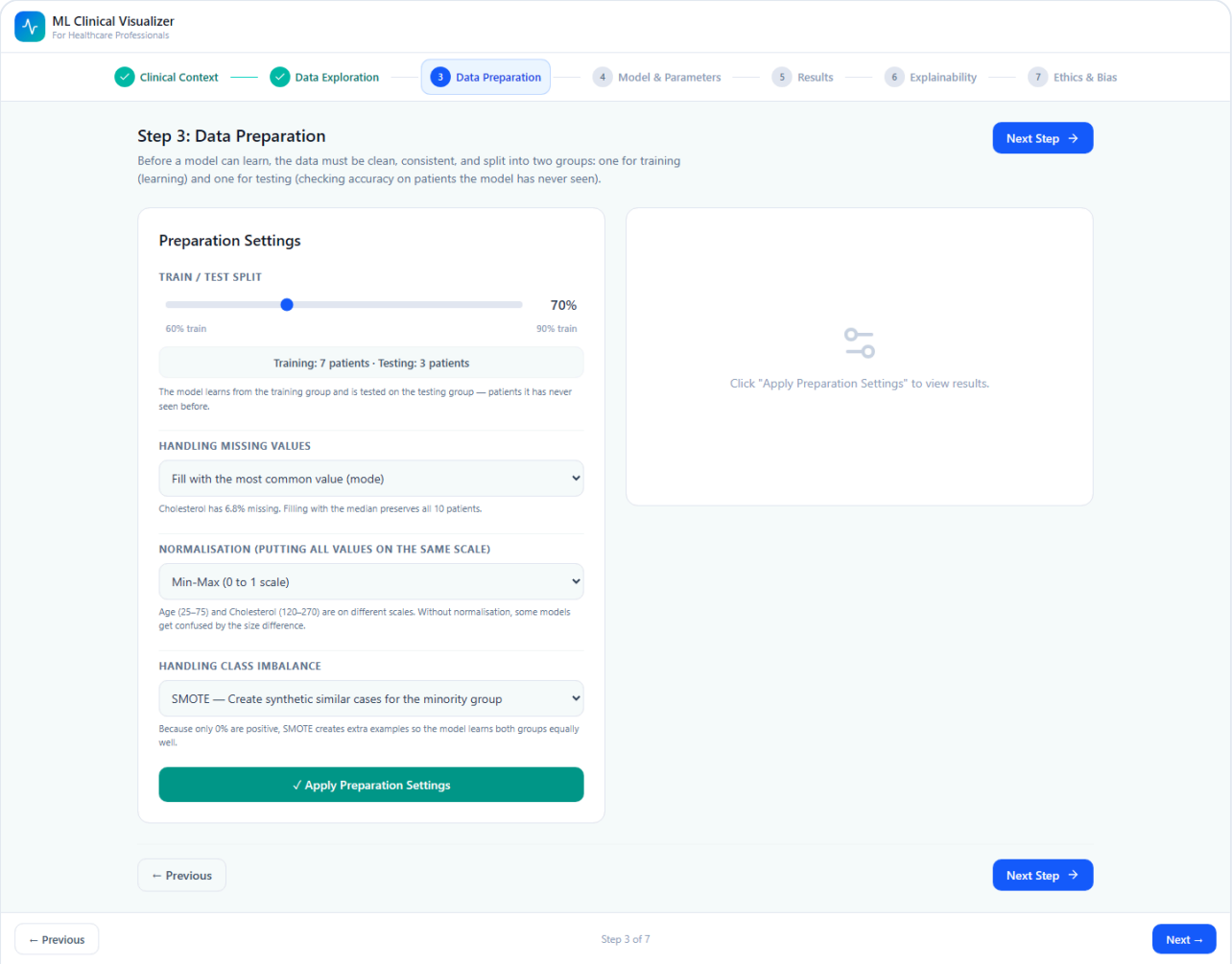
Step 3A: Controls visible (slider + dropdowns)



Screenshot: step3/controls_all_visible.png

The train/test split slider and preparation controls (missing values, normalisation, SMOTE/class-balance) are visible, satisfying the Step 3 UI control visibility requirement.

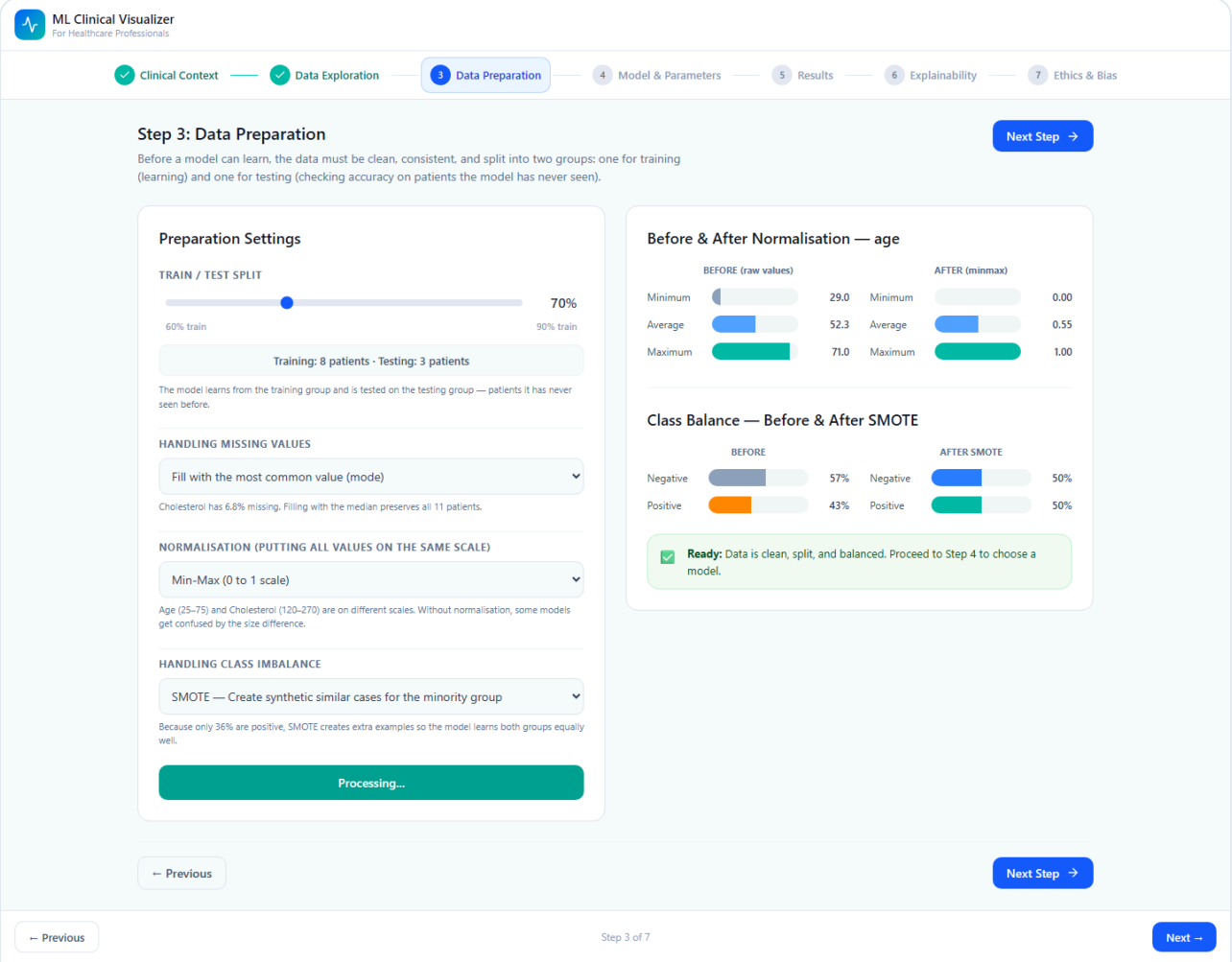
Step 3B: Controls in use (options selected)



Screenshot: step3/controls_selected_options.png

Controls are actively used: train/test split adjusted and options selected for missing values, normalisation, and SMOTE/class balance—demonstrating interactive configuration.

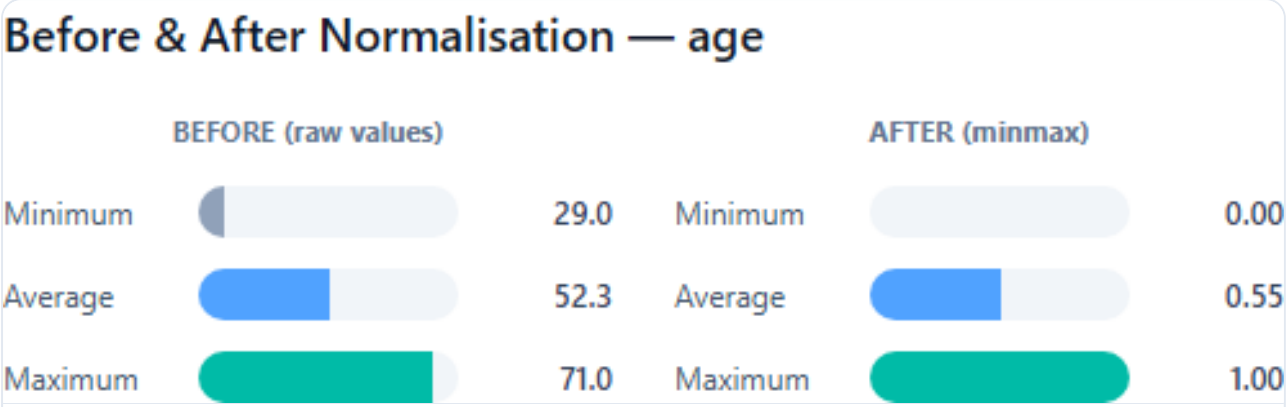
Step 3C: Apply clicked (processing triggered)



Screenshot: step3/processing_triggered.png

Clicking “Apply Preparation Settings” triggers a visible processing state, proving preprocessing is executed (not just UI-only).

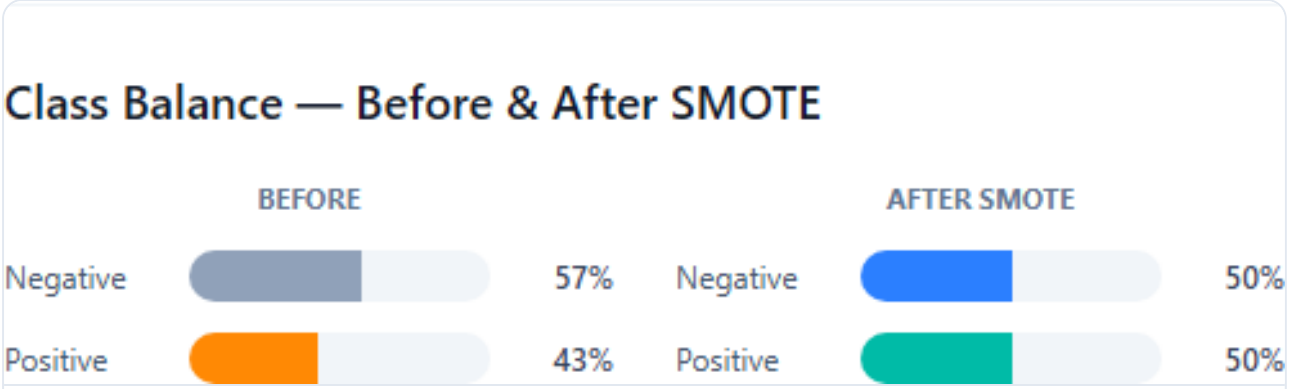
Step 3D: Before/after normalisation displayed



Screenshot: step3/before_after_normalisation.png

Before/after normalisation charts are displayed, showing the effect of the selected normalisation method (e.g., Min-Max vs Z-score), meeting the Step 3 chart requirement.

Step 3E: Before/after SMOTE class balance displayed



Screenshot: step3/before_after_smote.png

Before/after class balance charts are displayed for SMOTE/class-balance. SMOTE is applied to the training split only, consistent with the user guide and best practice.

Step 3F: Green success banner after apply

Ready: Data is clean, split, and balanced. Proceed to Step 4 to choose a model.

Screenshot: step3/apply_success_ready_banner.png

After preprocessing finishes, a green “Ready” success banner confirms preparation completed successfully and the pipeline can proceed.